

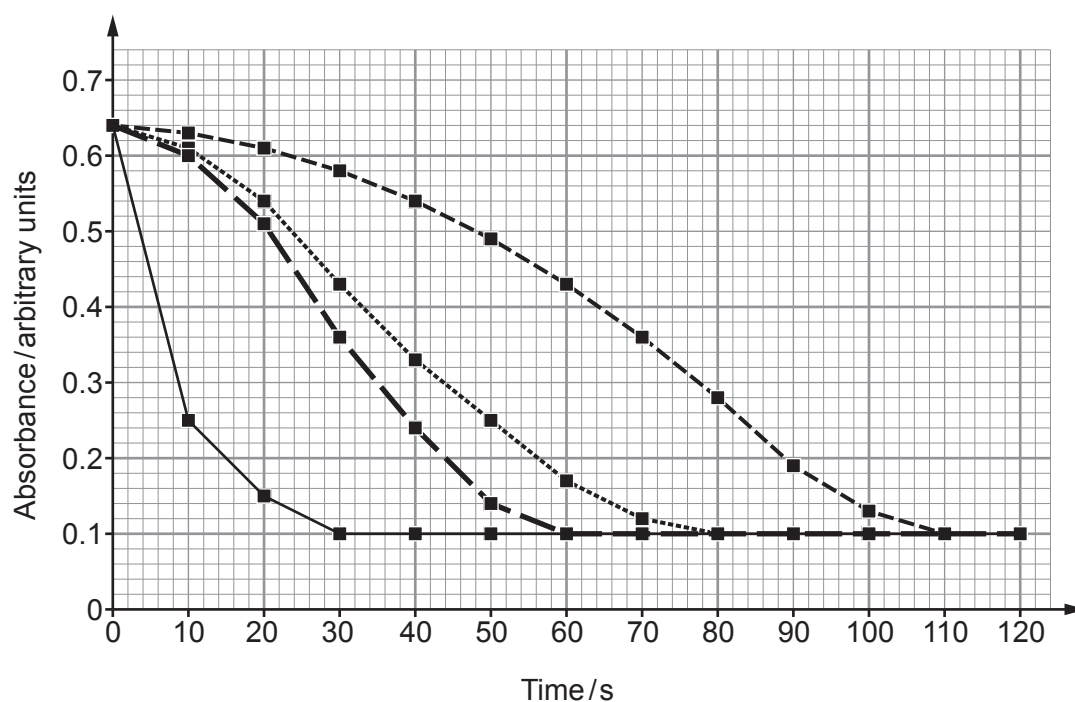
9. Chloe was investigating the effect of using catalysts on the rate of reaction.

She added 50 cm^3 of 0.1 mol dm^{-3} iron(III) nitrate solution to 50 cm^3 of 0.2 mol dm^{-3} sodium thiosulfate solution. The reaction forms a deep violet iron(III) complex which is unstable and is gradually reduced to form a light green iron(II) complex.

Chloe monitored the rate of reaction by measuring the absorption of light at a wavelength of 500 nm every 10 seconds for two minutes using a data logger.

- (a) The violet complex appears black at the beginning of the reaction. State the name of the technique used to monitor the rate of reaction by measuring the absorption of light. [1]

- (b) Chloe repeated the experiment three times adding 1 cm^3 of a different catalyst each time at a concentration 0.10 mol dm^{-3} . Below is a graph showing her results:



Key: - - - - - $\blacksquare$ - - - - - No catalyst - - - - - $\blacksquare$ - - - - - Fe^{2+}
 ······ $\blacksquare$ ······ Co^{2+} ——— $\blacksquare$ ——— Cu^{2+}

- (i) State which catalyst is the most effective. Explain your answer. [2]

.....

.....

.....



- (ii) Calculate the initial rate of reaction for the reaction catalysed by the copper(II) ions. [2]

rate = s^{-1}

- (iii) Each catalysed reaction contained the same number of moles of catalyst at the beginning of the reaction. Calculate the moles of catalyst left at the end of the reaction. [1]

moles = mol



- (c) Increasing temperature and the addition of a catalyst are two ways of increasing the rate of a reaction.

Using your knowledge of the Boltzmann distribution and particle theory, explain how the rate of reaction is increased using these two different methods.

You may use a diagram(s) to support your answer.

[6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

